

Stretchy: Enhancing Object Sensation Through Multisensory Feedback and Muscle Input

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ABSTRACT

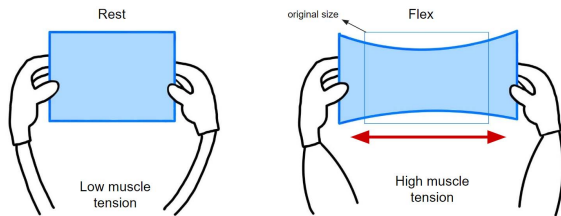


Figure 1: A stretchable virtual fabric depends on the user's strength of muscle detected by EMG sensors. The fabric patch before being stretched (left). The fabric patch being stretched under muscle tension (right)

Current works on 3D interaction methods mainly focus on rigid object manipulation and selection, while very few have been done on elastic object interaction. Therefore, we suggest a novel interaction method to observe and manipulate virtual fabric in a VR environment. We use multi-sensory pseudo-haptic feedback (a combination of tactile and visual feedback) and muscle strength data (EMG) to perceive the stiffness of the virtual fabric and flexible objects. For demonstration, we make fabric patches with various stiffness, and the stiffness difference can be distinguished. Our system can be implemented in a virtual cloth store to give consumers information about product stiffness and texture.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—User centered design;

1 INTRODUCTION

To create a realistic VR interaction, the implementation of haptic feedback is necessary to sense the object's properties. While many works have been done on weight and mass distribution perception, the stiffness sensation of soft, thin, and flexible materials such as fabric is still lacking. Therefore, we suggest a novel approach to provide a realistic 3D interaction between the user and the virtual fabric. In this paper, we generate the perception of stiffness towards the virtual fabric in the VR environment by combining multi-sensory feedback and muscle strength input. The user's hand position and

the edge of the fabric will be displayed as a virtual offset with tactile feedback as a guideline for the user to make correct hand placement. Our work makes the following contributions:

- Generating the perception of stiffness towards the virtual fabric in VR by combining multi-sensory feedback and muscle input.
- Suggesting a novel interaction method to enhance the realistic sensation of objects with elasticity in VR.

2 RELATED WORKS

As applications of XR technologies are growing rapidly, various studies are aiming to improve immersion by rendering a sense of physical experience to virtual objects through haptic cues. To render those sensations, one of the common approaches is employing wearable haptic devices. However, in some cases, natural haptic feedback can be complex due to hardware limitations. One alternative method is to use pseudo-haptic feedback as a replacement for a haptic cue. Pseudo-haptic feedback is using other sensory (visual, auditory, and somatosensory) as feedback to simulate the haptic experience. Visual is the most popularly used sensory to imitate haptic experience.

Visual Feedback : Many works use visual manipulation such as visual illustration, and visual tracking offset. For example, Rietzler et al. [2] simulate weight using perceivable tracking offset between the user's hand position and virtual object position (eg. the heavier object, the larger offset, and the slower movement). Despite of no additional hardware requirement, only visual feedback is provided and the revelation between the user's input and object property is relatively low.

Multi-sensory Feedback : To enhance sensation, Kim et al. [1] applies multi modals feedback by combining Control to Display ratio manipulation and EMS on stimulated weight perception. The disadvantage of EMS is that each muscle has a different sensation which is also different for each people make it hard to generalize. While Rietzler et al. [3] also deals with muscle activity, rather than electrically stimulating the muscle, they encourage the user to tense a muscle to achieve the desired interaction. In their work, based on the magnitude of muscle activity measured by the EMG sensor and the required threshold for interaction, the visual offset is displayed to guide whether to input force is too low or too high. However, with visual feedback and muscle input, some users needed time to get familiar with the system. To address this issue, we employ tactile feedback to assist users to make correct hand movements.

3 METHOD

Muscle Activity Input: Our approach will treat the stiffness of a fabric patch as an elastic spring corresponding to Hook's Law ($F = kx$). The stiffness of each type of fabric will depend on the k value. Two opposite virtual forces due to the fabric's stiffness F_k are applied when the user stretches the virtual fabric patch. The muscle force F_m muscle force acts in the opposite direction of the F_k and its magnitude depends on the measured muscle tension. The

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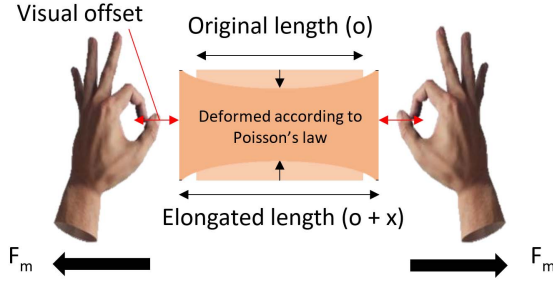


Figure 2: Implementation concept

fabric will elongate at x distance according to the $F_m = kx$ equation as the user stretch. Two modules of Myoware are used to measure muscle activity. Each module is attached to each arm of the user as both arms' muscles are tensed during stretching. Both modules are connected to Teensy 4.0.

Before the user test, calibration must be done first as the EMG signal varies between users. The reading data from Myoware are streamed to the Unity via serial IO port Lastly, convert the EMG values to force by multiplying the values by 110 N [3].

$$Force(N) = \frac{Myodata - min}{max} \times 110$$

where min and max are the minimum and maximum values that can be sent from Myoware sensors (the range is 0 to 1023 but also varies for each individual) while Myodata is the measured value of the user's muscle tension. For simplicity, Myodata obtains from the average tension of both arms.

The scaling factor is computed and will be used as a value of the max scale constraint which determines the maximum length that the fabric can extend according to the input force.

$$S(Scaling\ factor) = \frac{F}{kO} + 1$$

where k and O are the stiffness and the original length of fabric respectively.

Bimanual Interaction: We are using Oculus2 to track hand movement. A new script is modified from the existing two-hand scaling function offered in the software development kit. Instead of scaling the object in every direction, only the axis that parallels to the movement of the hands will be scaled.



Figure 3: User's arms with Myoware sensors (left), Demonstration (right)

4 DEMONSTRATION

For demonstration, we test our interaction method with three different fabrics with identical sizes (20 cm length and 10 cm width) but different stiffness. There are knit, cotton, and denim and the assigned stiffness value are 100N/m (low), 150N/m (medium), and 300 N/m (high) respectively. The muscle strength was maintained

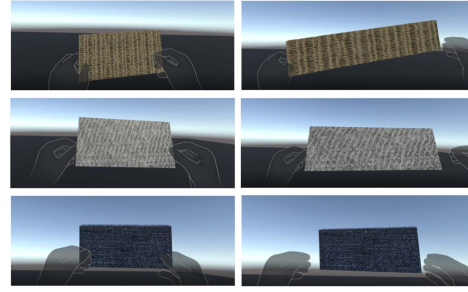


Figure 4: Stretching patches: Knit with low stiffness (top), Cotton with medium stiffness (center), Denim with high stiffness (bottom)

at around 15N. As shown in the picture, each fabric is stretched and elongated into different lengths. High-stiffness materials such as denim patches are harder to stretch, therefore more force is required to elongate while softer fabric with low stiffness such as knit patches is easy to extend. Tactile gloves provide the guidance when user's hands are not touching the object (stop vibrating) which helps indicate when the user is trying to stretch the patch more than the maximum extended length under the detected muscle tension. In this case, the identical vibration amplitude, and frequency are applied for each type of fabric patch.

5 CONCLUSION

We have suggested and implemented a novel interaction approach to simulate the stiffness of virtual objects. Our method is not only applicable to virtual fabric but can also be applied to other virtual objects. Currently, the fabric patch still lacks realism as it is made with a rigid plane object so in future work the clothes' physics properties should be applied. Moreover, as each object has a different surface characteristic, we are planning to adjust the vibrating frequency according to the object's texture for a better haptic experience. We believe the potential application of our work is to be utilized as an interaction tool in the online virtual clothes store. Since stiffness is an important property to characterize the impression of the fabric, users can obtain more information about the product compared to the existing virtual clothes shop.

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